MIT | Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	COURSE SYLLABUS	
DEPARTMENT OF Computer Science and Engineering(AIML)	**W.E.F**	**AY:** 2025 - 2026 (Rev. 2023 (NEP))
SECOND YEAR BACHELOR OF TECHNOLOGY CSE(AIML)	**COURSE NAME**	OPERATING SYSTEM
	COURSE CODE	2311213T
	COURSE CREDITS	3
RELEASE DATE : 01/07/2025	**REVISION NO.**	0.0

TEACHING SCHEME (HOURS/WEEK)		**EXAMINATION SCHEME AND MARKS**					
		THEORY			LABORATORY		TOTAL
LECTURE	PRACTICAL	IA	MSE	ESE	CA	PRACT/DEMO/PRES.	
3	NIL	30	20	50	NIL	NIL	100

PREREQUISITE KNOWLEDGE: Programming Knowledge

COURSE OBJECTIVES:

2311213T.CEO.1: To impart knowledge of key operating system concepts and its services.
2311213T.CEO.2: To provide comprehensive understanding of the concepts related to process, Inter process communication (IPC), scheduling and deadlock management
2311213T.CEO.3: To discuss memory management schemes in operating system.
2311213T.CEO.4: To develop shell scripting skills to automate routine operating system tasks and system-level operations.

COURSE OUTCOMES:

After successful completion of the course, students will be able to,

2311213T.CO.1: Employ OS concepts of structure, types, and functions to solve basic system-level problems. [L3]
2311213T.CO.2: Apply process management, memory management, and synchronization techniques to solve system-level problems.[L3]
2311213T.CO.3: Implement shell scripting techniques for process automation in operating systems[L3]
2311213T.CO.4: Analyze file systems, I/O management, and scheduling algorithms to improve system efficiency[L4]
2311213T.CO.5: Analyze deadlock conditions and propose strategies for prevention and avoidance in systems.[L4]

COURSE ABSTRACT:

This course provides a comprehensive introduction to the design, structure, and functionality of modern operating systems. It covers fundamental concepts such as process management, memory management, file systems, I/O systems, and concurrency control. Students will explore how operating systems serve as an interface between hardware and user-level applications, ensuring efficient resource allocation, system security, and user convenience. Through practical labs and case studies, learners will gain hands-on experience with process scheduling, deadlock handling, virtual memory, and system calls. The course also includes contextual modules that demonstrate OS principles in emerging fields like cloud computing, mobile systems, and AI/ML workload management.

THEORY COURSE CONTENT

UNIT 1	**Introduction to Operating System**	**5 HOURS**

App/System/Case study: Open Source Operating System-Linx,Role of Ubuntu in TensorFlow-based ML development **Contents:**
Definition, Types, Objectives, and Functions of OS,System Calls, Kernel/User Mode, OS Structures,OS support for AI/ML applications: multitasking, GPU control, and memory management
Self -Study: Microkernel vs Monolithic kernel design
Further Reading: Use of OS in various computing environments

UNIT 2	**Process and Thread Management**	**8 HOURS**

App/System/Case study: Programming Language multiprocessing in AI model training **Contents:** Process Lifecycle and Process Control Block, Threads and Multithreading Models, CPU Scheduling: FCFS, SJF, Round Robin, Priority, Context Switching, Process vs Thread in AI execution
Self-Study: Multi-threaded vs multi-process architecture in ML pipelines
Further Reading: Operating System Debugging

UNIT 3	**Process Synchronization and Deadlocks**	**8 HOURS**

App/System/Case study: Multithreading in Web Servers
Contents: Race Conditions, Critical Section Problem,Synchronization Mechanisms: Semaphores, Mutexes, Monitors, Deadlock: Detection, Prevention, Avoidance
Self-study: Locks and atomic operations in PyTorch/TensorFlow
Further Reading: Scheduling Algorithms in RTOS,Python concurrency models

UNIT 4	**Memory Management**	**8 HOURS**

App/System/Case study: Memory management issues in large model training
Contents: Memory Hierarchy, Address Binding,Paging, Segmentation, Virtual Memory,Page Replacement Algorithms,OS memory strategies for large ML models
Self- Study: System Programming with Linux
Further Reading: Use of memory profilers in Python

UNIT 5	**File and I/O Management**	**8 HOURS**

App/System/Case study: ile I/O bottlenecks in real-time inference systems
Contents: File Systems, File Allocation Methods, Directory Structures, File Operations,I/O Buffering, Device Drivers, Disk Scheduling, I/O efficiency in AI data loading pipelines
Self- Study: async file operations
Further Reading: I/O Optimization.

UNIT 6	Linux Essentials and Shell Programming	8 HOURS

App/System/Case study: Bash script to automate model
Contents: Linux File System Structure, Users Permissions,Shell Scripting: Variables, Loops, Conditions, Functions,Process Automation Cron Jobs,Use of Shell scripts in ML data pre-processing and deployment
Self- Study: Writing portable shell scripts using bash, awk, sed
Further Reading: GitHub: DevOps scripting repositories

TEXT BOOKS

1. Silberschatz, A., Galvin, P. B., Gagne, G. (2023). Operating System Concepts (10th Edition, International Student Version). Wiley. ISBN: 978-1119800361
2. Tanenbaum, A. S., Bos, H. (2023).Modern Operating Systems (5th ed.). Pearson Education.ISBN: 978-0137619832
3. Crowley, C. (1998). Operating systems : a design-oriented approach. Tata Mcgraw-Hill. ISBN : 978-0074635513

REFERENCE BOOKS

1. Stallings, W. (2018). Operating systems : internals and design principles. Pearson. ISBN-13: 9780137516742
2. Deitel, H. M., Deitel, P. J., and Choffnes, D. R. (2004). Operating Systems. Prentice Hall. ISBN-13:978-0131828278
3. Bovet, D. P., a Cesati, M. (2021 Reprint).Understanding the Linux Kernel (3rd ed.). O'Reilly Media.ISBN: 978-9352136342
4. Love, R. (2021).Linux Kernel Development (4th ed.). Addison-Wesley Professional.ISBN: 978-0134167398

REFERENCE LINKS

1. Linux Kernel Documentation (latest) Updated official Linux kernel reference. https://www.kernel.org/doc/html/latest/
2. MIT Operating Systems Engineering (6.828)-Advanced operating system design resources. https://pdos.csail.mit.edu/6.828/
3. Operating Systems: Three Easy Pieces – Free PDF, Modern, student-friendly OS textbook with labs and case studies. https://pages.cs.wisc.edu/ remzi/OSTEP/
4. LinuxCommand.org, Great for beginners learning shell scripting and CLI basics.